Noise tracking for Wiener speech enhancement

Hard gating and soft speech presence on NOIZEUS

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Project website and demonstration materials

Repository and downloadable code

Video presentation and code demonstration

Supporting benchmark and sentence sensitivity results

Abstract

This project tests how noise estimation affects a two-step Wiener speech enhancer. A shared processing path compares fixed estimation, hard energy gates and a literature-based soft speech-presence probability (SPP) tracker. A development-selected gate improves mean STOI by 0.00404 over the original gate on 48 unused sentence conditions, but remains below fixed estimation. Matched noise-level changes then expose blocked updates after a rise. A frozen follow-up tests SPP on 216 new mixtures from 18 additional sentences by the same six speakers, with 36 earlier mixtures kept separate for regression. After a +6 dB rise, source-default SPP reduces mean late four-band power error by 5.762 dB and adds 1.593 dB reference SNR relative to the selected gate, but loses 0.01398 STOI. A predeclared slower-rate ablation reduces that STOI loss to 0.00252 while recovering less power-tracking accuracy. The experiments identify an update blockage and test a mechanism-based response. Improved noise tracking does not establish better intelligibility or perceptual preference.

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1 Research question and contribution

Problem and motivation

Single-channel speech enhancement must estimate speech and noise from their mixture. Reducing background energy can also remove weak speech. A useful investigation therefore needs a reference signal, complementary metrics and an explanation of the conditions in which the processing helps or fails.

A Wiener enhancer uses a noise power spectral density (PSD) estimate in every frequency bin. Holding the initial estimate fixed is stable but cannot track later changes. Updating during detected pauses can track noise if suitable frames are admitted. An overestimated noise PSD tends to lower the estimated speech-to-noise ratio and gain, risking weak-speech suppression; an underestimate can leave more residual noise. Weak speech admitted as noise and loud noise mistaken for speech create different estimation failures.

Research question

When does noise updating improve a TSNR Wiener enhancer compared with a fixed initial estimate? Can soft speech-presence weighting recover from a hard gate's blocked updates without sacrificing predicted intelligibility? The shared enhancement path isolates the noise tracker rather than ranking complete speech enhancement systems.

Contribution

The TSNR core derives from LIU Ming and Pascal Scalart [6] and Plapous et al. [1]. The project contribution is the integration and controlled investigation of noise-update policies. A common amplitude-preserving reconstruction supports reference scoring. Section 5 selects a protective energy gate on development data. Sections 6 and 7 impose matched level changes and identify its tracking failure. Sections 8 and 9 implement Gerkmann and Hendriks' SPP estimator [12] within the same TSNR path, then test recovery and speech cost on additional sentences. Neither TSNR nor SPP is claimed as a new algorithm.

The analysis uses sample-aligned clean and noisy NOIZEUS recordings [2,5]. All methods receive the same noisy signal and use the same window, initialisation, gain calculation and reconstruction. Clean speech is used for mixture construction, scoring and diagnostics, never as an input to the enhancer. The original data, source licence and numerical outputs remain available in the accompanying code package.

What the experiments establish

The protective gate improves mean STOI relative to the original rule but admits few updates. Controlled rising noise can therefore remain underestimated. Soft, frequency-wise weighting reduces this tracking error and raises reference SNR, while its mean STOI falls in the primary rising-noise comparison. A predeclared smoothing-rate ablation exposes the trade-off between recovery and speech protection. The evidence supports a mechanism-based improvement in tracking, not a universally superior enhancer.

The report distinguishes development, new-sentence evaluation, reused diagnostic data and the later frozen confirmation study. Its 18 additional sentences were unused in the earlier local experiments but come from the same six NOIZEUS speakers. Repeated conditions increase coverage without creating independent speakers. Original benchmark spectra and sentence-sensitivity details remain in the linked supporting results; their original numbering is retained.

2 Literature and design rationale

The enhancement path

Scalart and Vieira Filho [7] make a priori SNR estimation central to speech enhancement: errors in estimated speech or noise power affect the Wiener gain. Plapous et al. [1] use TSNR to refine the delayed decision-directed estimate. Their harmonic-regeneration extension is not implemented here. Keeping TSNR fixed isolates the effect of the noise-update policy.

Choosing a noise estimator

Sohn et al. [8] use likelihood-ratio decisions and temporal modelling for statistical VAD. The present broadband energy gate uses less structure, but its threshold and hangover expose the trade-off between protecting speech and admitting updates. It does not reproduce the Sohn detector.

Martin [9] tracks smoothed spectral minima; MCRA [4] and improved MCRA [10] regulate averaging using speech-presence information. Gerkmann and Hendriks [12] use a soft speech-presence probability with fixed priors. Their binwise weighting can retain update opportunities that a broadband binary decision blocks. This motivates the implemented follow-up in Section 8. Minima-based estimators are discussed but not implemented; SPP is tested with a prespecified smoothing-rate ablation.

Estimator	Useful property	Principal limitation here
Fixed initial PSD	Stable common baseline	Cannot follow later changes
Energy VAD updating	Simple decisions and explicit update times	Speech contamination or blocked tracking
Soft probability [12]	Frequency-wise graded updates	Can admit speech or recover slowly

The hard gate provides inspectable update decisions. The matched level changes expose a concrete failure, then the SPP comparison tests whether changing only the tracker addresses it. The two frozen SPP rates distinguish the effect of soft weighting from that of faster PSD smoothing.

Relation to recent enhancement systems

DeepFilterNet [13] combines learned ERB-band gains for the spectral envelope with complex multi-frame filtering for speech periodicity, demonstrated on 48 kHz audio. Its learned representation and reconstruction differ from the present 8 kHz TSNR pipeline. A direct comparison would assess complete systems rather than isolate noise updating. The present study therefore provides an interpretable component investigation; it does not establish a ranking against learned models or claims about their robustness.

What the measurements mean

Hu and Loizou [2] distinguish speech distortion, background-noise distortion and overall quality. Taal et al. [3] introduce STOI as an intelligibility predictor for noisy and time-frequency processed speech. Reference SNR and STOI therefore provide complementary evidence; STOI is neither a word-recognition percentage nor a listening-quality score.

Two distinctions guide the experiment: the noise PSD describes the estimated background, whereas a priori SNR controls attenuation using that estimate; a lower waveform error need not imply higher intelligibility. Paired metrics and update traces test these different aspects. References [5,6,11] identify the data and software used.

3 Processing method

Common TSNR analysis and synthesis

For additive, uncorrelated speech and noise, the minimum-mean-square-error linear spectral gain is speech power divided by total power, equivalently ξ / (1 + ξ) for a priori speech-to-noise ratio ξ [7]. The implemented gains use estimated powers and additional constraints [1], so the ideal model does not guarantee minimum error on each recording.

The input is mono at 8 kHz. A 200-sample window, 80-sample hop and 400-point FFT implement 25 ms analysis and 10 ms updates. The 20 Hz FFT-bin spacing includes zero padding; the 25 ms window and its shape determine frequency resolving power. The source Hann convention is w[n] = 0.5 - 0.5 cos(2 pi n / 201), n = 1,...,200. All methods share these settings.

Let $Y_t[k]$ denote a noisy-frame spectrum and $D_t[k]$ the current noise PSD. The source branch forms a floored instantaneous estimate $q_t = \max(|Y_t|^2 / D_t - 1, 0.1)$. The decision-directed estimate combines q_t with the previous enhanced magnitude using a 0.99 weight on the previous estimate. The following expressions apply separately in each frequency bin:

$$\xi_t = 0.99\ |\hat{S}_{t-1}|^2\ / D_t + 0.01\, q_t$$

$$G_{1,t} = \xi_t / \left(1 + \xi_t\right); \xi_{2,t} = |G_{1,t} Y_t|^2\ / D_t$$

A second Wiener gain $\xi_{2,t} / (1 + \xi_{2,t})$ has a pre-constraint floor of 0.15. The inherited constraint [6] inverse-transforms this gain, retains 200 impulse-response samples around zero lag with Hamming weighting, and takes the FFT magnitude. Rescaling restores the pre-constraint mean-square gain; 0.15 need not bound the final gain. This common operation applies an impulse-response duration constraint without testing a new gain design. Observed phase is retained, with identical reconstruction in all modes.

Noise initialisation and updating

Every mode starts from the same mean power spectrum of 601 windows advanced one sample at a time within the first 0.10 s. Fixed estimation retains this PSD. The gate computes a broadband energy ratio, with each sum taken over the FFT bins:

$$R_t = 10 \log_{10}\left(\Sigma\ |Y_t[k]|^2\ /\ \Sigma D_t[k]\right)$$

A ratio R_t greater than threshold θ declares speech and resets the hangover to H frames. The next H frames at or below threshold remain protected; another detection restarts the count. Protected frames retain D_t. Eligible unprotected frames update only after current-frame enhancement:

$$D_{t+1} = 0.95\, D_t + 0.05\ |Y_t|^2$$

The original gate uses 3 dB and five hangover frames. The selected gate uses 1 dB and ten frames. A lower threshold declares speech more readily; a longer hangover retains protection after a detection. Calibration and padded frames never update. Their clean reference is never an algorithm input.

Reconstruction and attribution

Independent peak normalisation is removed because it alters reference-error comparisons. With no additional synthesis window, overlap-add is divided by the shifted analysis-window sum, not its squared sum. Unity-gain testing validates amplitude and alignment, not enhancement quality. A separate legacy mode reproduces the source. Its licence and headers credit the inherited TSNR implementation [6]; reconstruction and update experiments are documented adaptations.

4 Experimental design and verification

Four experimental stages

Stage	Material and selection	Question
Original benchmark	2 development and 4 evaluation sentences; car/street at 0/5/10/15 dB	Does the original gate help fixed TSNR?
Parameter study	9 settings on 16 development cases; 48 cases from 6 unused sentences	Does a more protective gate improve the original rule?
Controlled diagnostic	Same 6 test sentences; constant/rise/fall profiles; 36 mixtures	How do imposed changes interact with the frozen gates?
SPP confirmation	18 additional sentences; 216 new mixtures; 36 old mixtures separate	Does soft weighting recover tracking, and at what cost?

The original evaluation spans four speakers other than the two development speakers. The later six sentences are sp02/07/12/17/22/27. The SPP confirmation adds the remaining three sentences per speaker: sp03–05, sp08–10, sp13–15, sp18–20, sp23–25 and sp28–30. The six speakers are known; the 18 sentences were unused. The follow-up protocol, parameters and example were recorded before execution, with no test-set selection.

One intelligibility implementation throughout

All main intelligibility scores use standard pystoi 0.4.1, extended=False [11], on unquantized outputs. Original outputs were retrospectively rescored without changing their waveforms. This corrects the count improving both SNR and STOI from six to five of 32. Across 144 archived scores, mean absolute implementation disagreement is 0.00285; it is not an error bar on within-pystoi comparisons.

Scoring and interpretation

Reference SNR is 10 log10 of clean energy divided by processed-minus-clean error energy. It penalises residual noise and speech distortion together. All whole-recording metrics exclude samples 1:800; there is no delay fitting, gain fitting or per-method amplitude normalisation. Nominal NOIZEUS SNR labels use active speech scaling [5] and need not equal this measured SNR. Means are arithmetic means of case scores, not pooled waveform SNR.

Each paired difference uses the same input and clean reference. The six-sentence test uses sentence-level leave-one-out means. In confirmation, constant-benchmark means average eight conditions per sentence; each controlled profile averages two noises. Three sentence means then form each of six speaker means. All conditions and failures are retained. These descriptive summaries are not population confidence intervals.

Diagnostic definitions

Valid frames start after sample 800 and exclude padding. Clean-frame energy at or above -30 dB relative to each recording's peak defines a speech proxy; at or below -40 dB defines a low-energy proxy. These diagnostic labels are never algorithm inputs. Hard-gate overlap divides updated speech-proxy frames by all speech-proxy frames. Soft trackers report effective update weights, which differ from binary update coverage.

Verification

Checks cover source reproduction, unity reconstruction, shared initial PSDs, recurrence and metric arithmetic. All 1,512 follow-up rows are retained; earlier baseline waveforms reproduce exactly. Independent calculations match SNR within 8×10^{-14} dB and noise-power/PSD recurrences within 6×10^{-15} relative error. MATLAB R2026a performs enhancement; Python computes STOI. Protocols and per-case outputs support reproduction.

5 Gate selection and benchmark evidence

In the original 32-case evaluation, the original gate improves both SNR and STOI over fixed estimation in only 5/32 cases. Mean STOI changes are -0.00717 in car and -0.00103 in street noise. This motivates a more protective gate. The 0.95 smoother is retained while thresholds 1/3/5 dB cross hangovers 0/5/10 frames on 16 development cases. The grid brackets 3 dB by ±2 dB and 50 ms by 0–100 ms. Mean development STOI selects the setting; ties within 10^{-6} use SNR, then proximity to the original settings. Test IDs and this rule were frozen beforehand.

Table 1 Development results and pooled speech-proxy update overlap

Threshold dB	Hangover frames	SNR dB	STOI	Overlap
1	0	11.467	0.7999	14.7%
1	5	11.375	0.8009	3.7%
1	10	11.300	0.8012	2.6%
3	0	11.256	0.7914	35.6%
3	5	11.265	0.7949	19.8%
3	10	11.227	0.7967	15.5%
5	0	10.670	0.7836	52.6%
5	5	10.748	0.7858	38.6%
5	10	10.566	0.7888	33.4%

At five-frame hangover, lowering the threshold from 3 to 1 dB adds 0.00598 STOI. At 1 dB, increasing hangover from five to ten adds only 0.00025 and reduces update coverage from 14.1% to 8.5%. The grid probes a local trade-off; it cannot establish a global optimum.

The selected 1 dB/ten-frame rule has development STOI 0.8012. The highest SNR occurs at 1 dB/zero hangover. The choice depends on the objective; the new test does not trigger retuning.

Table 2 Means over 48 new sentence conditions

Method	SNR dB	STOI	Proxy overlap
Input	7.197	0.8102	-
Fixed	10.651	0.8142	-
Original VAD	10.656	0.8095	16.0%
Selected VAD	10.715	0.8136	2.7%

Selected VAD adds 0.059 dB SNR and 0.00404 STOI over original VAD, but remains 0.00064 below fixed STOI. Proxy overlap falls from 16.0% to 2.7%; update coverage falls from 27.4% to 4.9%. Five of six sentence means favour selected over original. Every leave-one-sentence-out mean is positive (+0.00300 to +0.00491); against fixed, all remain negative (-0.00132 to -0.00007). These sensitivity checks and the detailed condition results are retained in the supporting pages.

6 Controlled changes in noise level

The earlier noise/SNR groups use different excerpts. This diagnostic instead keeps each underlying speech and noise pair identical, then changes only the noise envelope. It reuses six test sentences and car/street mixtures at nominal 10 dB. Subtracting aligned clean speech s[t] from the noisy source gives residual r[t]. The new input is y[t] = s[t] + a[t]r[t], with no gain fitting. The residual includes any source quantisation discrepancy.

The first 100 ms and the entire pre-change prefix are unchanged. Halfway through the remaining samples, a 50 ms ramp linear in dB reaches 0, +6 or -6 dB noise amplitude level, then holds. The 36 mixtures retain an unprocessed reference and use the frozen fixed, original and selected methods. This tests prescribed trajectories, not variability at equal average SNR. All parameters and the sp02/street/rise example were recorded before this run.

Table 3 Selected minus fixed: six sentence means, each averaging car and street

Noise profile	Whole SNR dB	Whole STOI	Late SNR dB
Constant	+0.0594	+0.001180	+0.0552
Rise +6 dB	-0.0359	+0.000914	-0.0578
Fall -6 dB	+0.1428	+0.002011	+0.3949

Whole scores exclude calibration. Late SNR begins 100 ms after the ramp ends and uses the same clean reference; short late segments are not scored with STOI. Figure 1 shows all six sentence effects and their means, without confidence intervals.

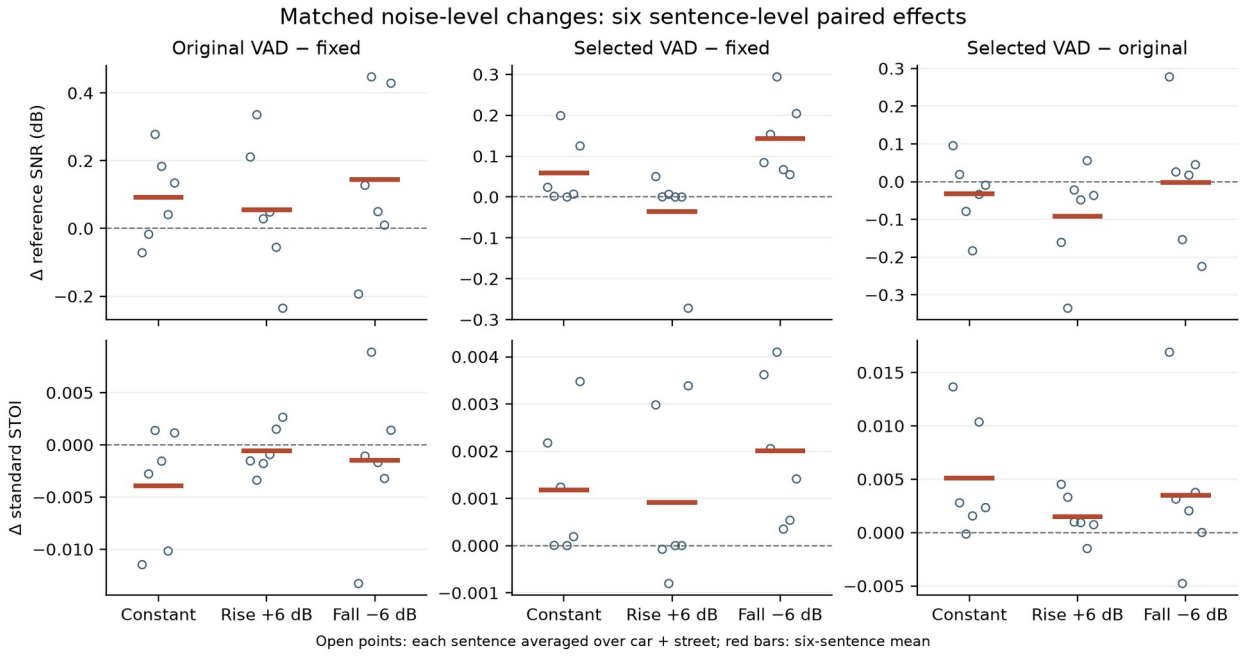


Figure 1 Matched profile comparisons. Open circles are sentence means over two noises; red bars average the six sentences. Axes differ across panels. The complete per-case results retain all outcomes.

Within the falling profile, selected VAD exceeds fixed SNR and STOI in all six sentence means, but its STOI of 0.9059 remains below the input's 0.9071. A matched profile contrast subtracts the constant-profile selected-minus-fixed difference from each changed-profile difference, averaging over two noises within sentence and then six sentences. The fall adds 0.08345 dB whole SNR, 0.000831 STOI and 0.33969 dB late SNR beyond the constant-profile advantage; its extra STOI gain is positive in five of six sentences. The rise changes these effects by -0.09526 dB, -0.000266 and -0.11303 dB, respectively. Thus Table 3 reports within-profile advantages, while these contrasts quantify their change from constant noise. They remain descriptive results for the imposed trajectories.

7 Hard gate tracking behaviour

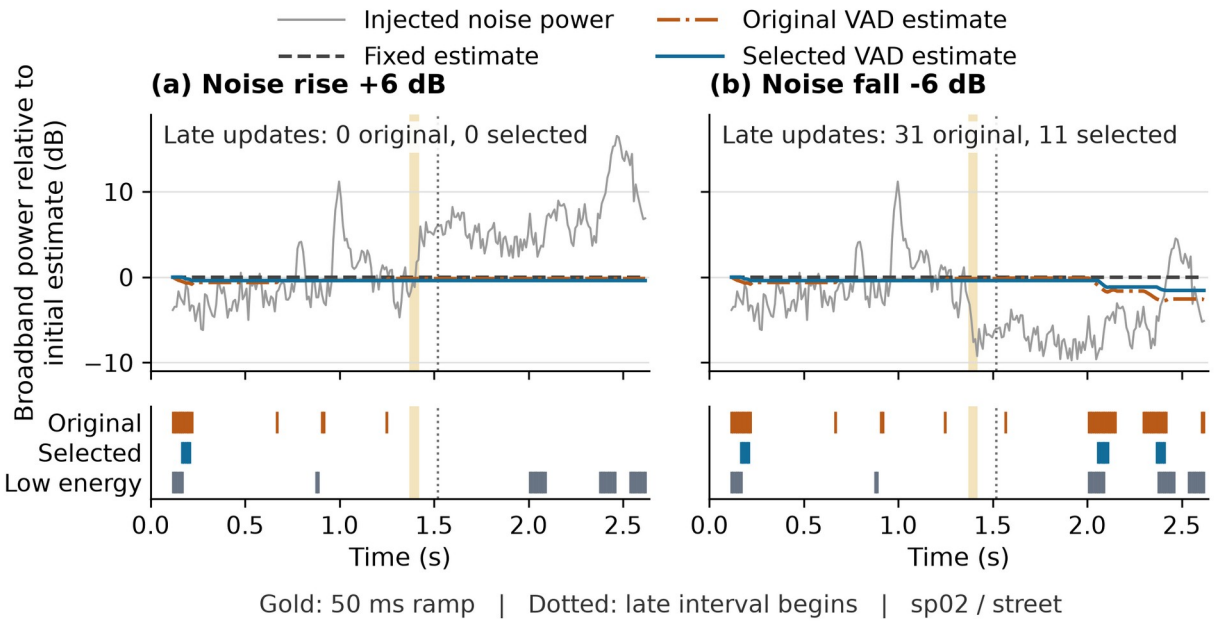


Figure 2 The original prespecified sp02/street rise example and its matched fall, using the same noise and speech. Estimates precede each update; all powers are relative to the shared initial estimate. Grey curves show windowed injected-noise power, not expected spectral PSD. Shading marks the ramp; dotted lines start the late interval. Raster marks show accepted updates and the low-clean-energy proxy, not manually labelled silence.

After the rise, both gates declare speech throughout all 109 late frames in this example, including 27 low-clean-energy proxy frames. Neither gate updates late in any of these 12 original rising-noise pairs. Mean per-case late estimated-to-injected broadband power ratios are -8.58 dB for original and -8.68 dB for selected VAD. A simple limiting case explains this blockage: with an accurate pre-rise estimate, a +6 dB noise-only power step gives R_t approximately 6 dB, above either threshold. The gate then labels noise as speech and retains the low estimate. Actual frame powers fluctuate; the saved traces establish the observed blockage.

Nonzero score differences do not demonstrate successful late tracking: the methods retain estimates and enhancement states established earlier, while whole-recording STOI also includes pre-change speech.

The matched fall admits 31 original and 11 selected late updates out of the same 109 frames. Across all 12 falling-noise pairs, mean per-case late update coverage is 34.7% and 17.6%, respectively. Mean late power ratios are +2.18 and +2.64 dB, versus +3.47 dB for fixed estimation. Lower noise permits updates, while the protective gate admits fewer. Each ratio compares summed estimated and injected powers within one late interval; the dB ratios are then averaged across pairs. They describe realised broadband power, not spectral accuracy.

Even with every frame admitted, a 0.95 smoother requires approximately ln(0.1)/ln(0.95) = 45 updates, or 0.45 s at a 10 ms hop, to remove 90% of an ideal constant-target initial error. Gating extends elapsed adaptation time; short speech recordings may end before settling. This calculation explains the possible time scale, not a fitted settling-time measurement.

The 12 constant-source inputs pass identity reconstruction; all 36 mixtures pass shared-prefix and initial-PSD checks. Independent Python calculations reproduce every diagnostic noise-power frame and the before/after update recurrence within 8×10^{-15} relative error. In matched late frames, a ±6 dB injection changes the fixed-estimator log ratio by exactly ∓ 6 dB. The constant-profile scores reproduce the prior 10 dB test, with maximum STOI difference below 5×10^{-16}.

8 Soft probability tracking

Replacing the blocked update rule

The follow-up adapts the fixed-prior SPP estimator of Gerkmann and Hendriks [12]. Each FFT bin receives a continuous speech-presence probability instead of the broadband hard decision. The TSNR gains, framing, 0.10 s initial PSD and reconstruction remain unchanged. With current periodogram P and pre-update noise estimate D, the speech prior is q = 0.5 and the active-speech model SNR is χ = 10^(15/10). This fixed 15 dB parameter is not TSNR's estimated a priori SNR.

$$\ell = \ln(q/(1-q)) - \ln(1+\chi) + \chi P/((1+\chi)D)$$

$$p = \exp(c)/(1+\exp(c)), \qquad c = min(\ell, 200)$$

Probability memory begins at m = 0.5 and updates as m ← 0.9m + 0.1p. If m > 0.99, p is capped at 0.99 to limit stagnation; memory uses the raw probability before this cap. After current-frame enhancement, the next noise estimate is

$$D_{\text{next}} = D + (1-\alpha)(1-p)(P-D).$$

The source-default α = 0.8 is the main comparator. A predeclared α = 0.95 ablation matches the hard gate's PSD smoother; neither is selected using confirmation scores. Source coefficients are retained on the common 10 ms hop, whereas the author example uses 16 ms. This is an adapted estimator, not a reproduction of that whole system. Noise PSD and probability memory freeze on calibration and padded frames. The effective observation weight is (1 − α)(1 − p).

Prespecified recovery endpoint

The primary endpoint is late four-band absolute log-power error after a +6 dB rise, versus selected VAD. Bands are 0–500, 500–1,000, 1,000–2,000 and 2,000–4,000 Hz. Within each band, sum estimated and injected-noise powers over identical late frames, take |10 log10(estimate/noise)|, then average the four errors. One-sided bins use weights 1 at DC/Nyquist and 2 elsewhere. This measures agreement with finite-window realised noise power, not expected PSD truth. Whole SNR and STOI assess output accuracy and predicted intelligibility.

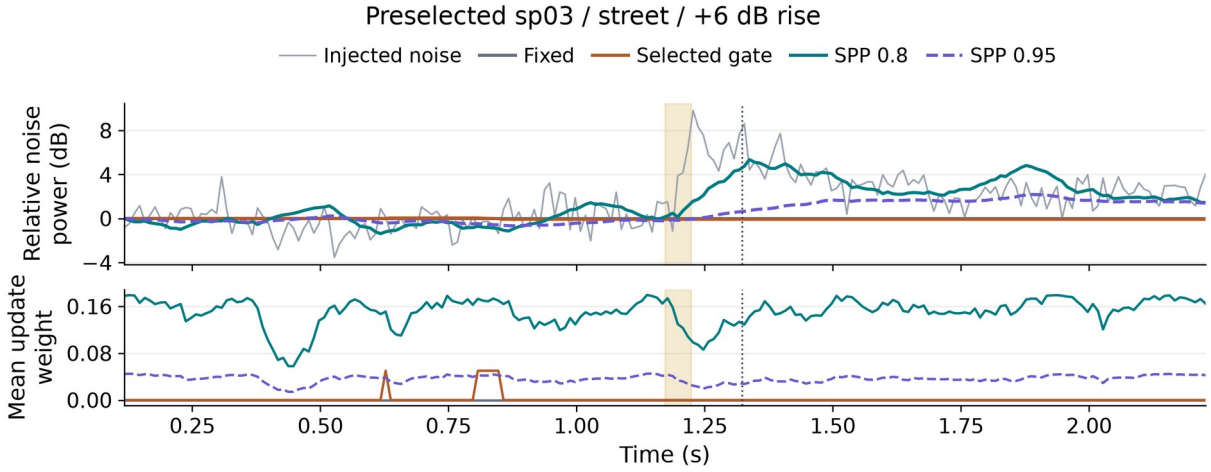


Figure 3 Prespecified sp03/street/10 dB rising-noise trace. Estimates precede updates; powers share the initial reference. Soft weighting permits recovery while selected VAD remains near its initial estimate. Aggregate intelligibility conclusions use Table 4 and Figure 4, not this example alone.

9 Confirmation on additional sentences

The frozen follow-up comprises 144 constant cases (18 sentences × 2 noises × 4 SNRs) plus 72 rise/fall cases at nominal 10 dB. The 36 earlier mixtures are diagnostic/regression data, reported separately. Six methods, including the unprocessed input, yield 1,512 rows in total. No settings change after results are seen.

Table 4 SPP minus selected VAD on 18 additional sentences

Profile	α	Δ band error dB ↓	Δ SNR dB ↑	Δ STOI ↑
Constant 0–15 dB	.80	-1.063	-0.059	-0.00780
Constant 0–15 dB	.95	-0.893	+0.199	-0.00390
Rise +6 dB	.80	-5.762	+1.593	-0.01398
Rise +6 dB	.95	-4.017	+1.382	-0.00252
Fall -6 dB	.80	-2.834	+0.131	-0.00058
Fall -6 dB	.95	-1.717	+0.288	+0.00010

Means first average conditions within sentence and then three sentences within each known speaker. Negative band-error changes favour SPP. The constant benchmark includes all four SNRs; Figure 4 instead holds nominal SNR at 10 dB for matched profile comparisons.

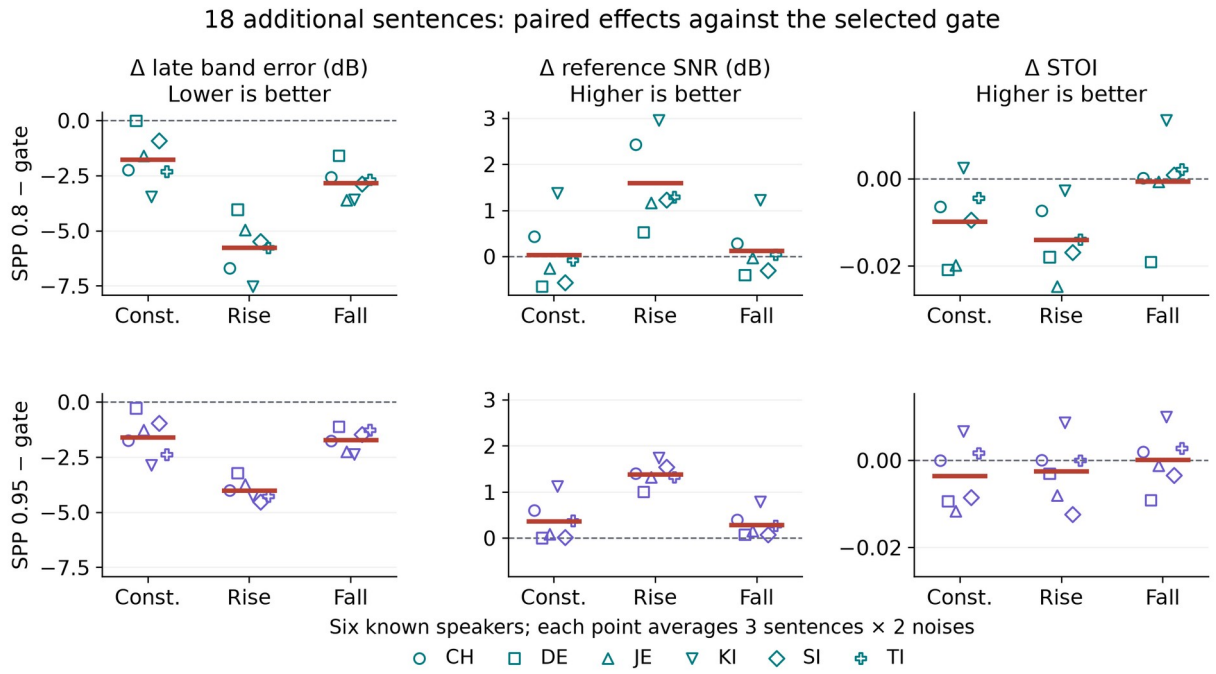


Figure 4 SPP minus selected VAD under matched 10 dB profiles. Points are six known-speaker means, each averaging three sentences and two noises; bars are their means. Different metric panels use different scales.

After a rise, default SPP reduces band error in all 18 sentence means and all six speaker means. Mean SNR rises by 1.593 dB, but STOI falls by 0.01398. The .95 ablation yields +1.382 dB SNR with -0.00252 STOI. Against fixed estimation in rising noise, default SPP gains 1.671 dB SNR but loses 0.01228 STOI. Selected VAD has zero late updates in 32/36 new rising-noise cases, not every case. Across all constant cases, default SPP also loses 0.00780 STOI versus selected VAD. Tracking gains therefore do not establish general enhancement superiority.

10 Discussion and conclusions

From gate protection to recovery

The first modification protects speech-associated frames: selected VAD improves average STOI over the original rule while cutting update coverage to 4.9%. Its small remaining STOI deficit against fixed estimation indicates that the gate does not justify a general replacement of that baseline. Controlled profiles then reveal why adaptation can fail. A rising noise level is labelled as speech, leaving the PSD too low; a fall admits updates. This mechanism motivates a change in estimator structure rather than another threshold search.

The frozen soft-tracker follow-up addresses that blockage on 18 additional sentences. After a rise, default SPP reduces mean late band-power error by 5.762 dB and increases reference SNR by 1.593 dB relative to selected VAD. However, mean STOI falls by 0.01398. The .95 rate ablation recovers less tracking accuracy but limits the STOI loss to 0.00252. Because both rates were specified before testing, these outcomes describe a trade-off; they are not grounds to declare an after-the-fact winning configuration.

The results give a concrete test of the probability-weighted alternative [12] motivated by the literature [4,9,10]. Continuous binwise weights can admit noise observations when a broadband gate is closed, but they can also admit speech. The higher speech-proxy update weights are consistent with this exposure, not a causal attribution of every STOI change. Faster $\alpha = 0.8$ smoothing improves recovery more strongly after a rise, whereas slower $\alpha = 0.95$ better preserves mean STOI. Constant-case losses show why a rise-only evaluation would have overstated success.

Limits of the evidence

The additional 18 sentences broaden lexical coverage to three new sentences per known speaker, but there are still only six speakers. Speaker means avoid counting repeated conditions as independent people; they do not establish performance on unseen speakers. The original six diagnostic sentences are kept separate. Short utterances constrain settling time, and the imposed envelopes change later SNR as well as temporal variation. The conclusions concern these trajectories, not variability at equal average SNR.

The clean-energy proxies are not human VAD annotations, and weak consonants can fall below them. Four-band error compares estimates with finite-window realised noise power, not an exact expected PSD. Better noise agreement can coexist with speech distortion. The first 0.10 s may contain weak speech, which contaminates the common initialisation. Source SPP coefficients also operate at a different hop from the author example; the comparison is fair within this shared TSNR path, not a whole-paper reproduction.

No subjective listening study or manual speech annotation was performed. STOI predicts intelligibility rather than listener preference, and the smaller score differences do not establish noticeable benefit. The experiments also leave the TSNR core unchanged and do not compare complete learned systems such as DeepFilterNet [13].

Conclusion and next investigation

The project identifies a hard-gate tracking failure, implements a literature-based response and tests it on additional sentences without retuning. Soft probability weighting improves recovery from an upward noise change, but the measured STOI cost prevents a claim of a generally better enhancer. The useful result is the quantified distinction between tracking accuracy, waveform error and predicted intelligibility, supported by common baselines and reproducible outputs.

The next investigation should use longer recordings and independently selected speakers, with development data to set a tracking-versus-distortion objective before an untouched test. Human speech annotations would check the proxy labels, and blinded listening would test perceptual benefit. Alternative probability calibration or minima-based tracking [4,9,10] should be compared within the same processing path. These are proposed tests, not completed results.

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